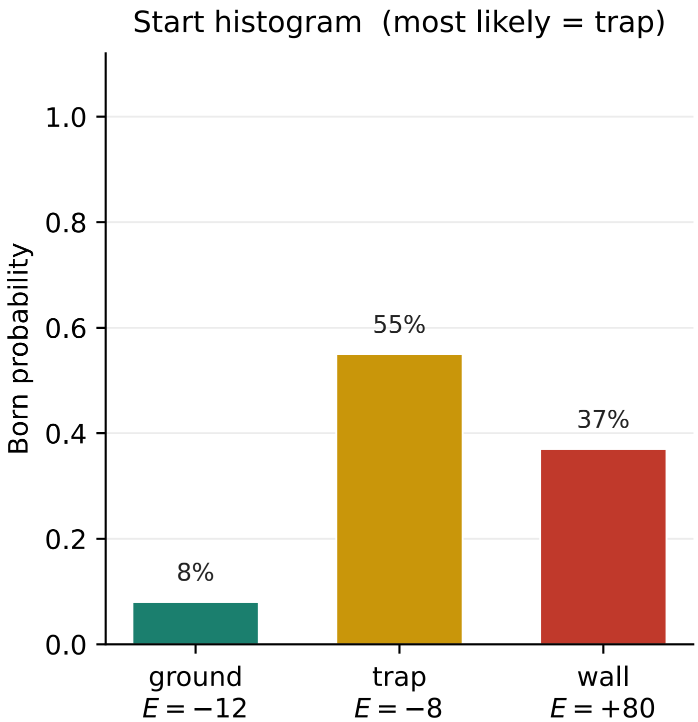
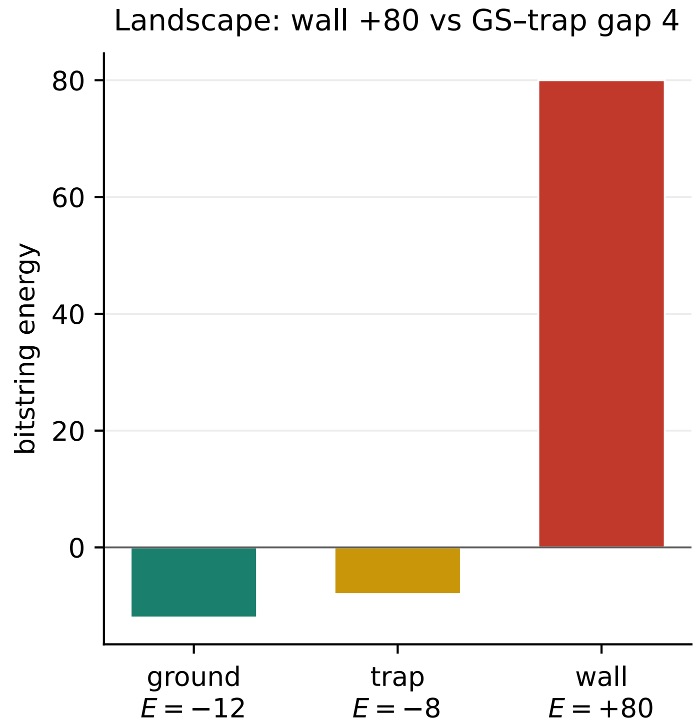
Why Gibbs finds the bitstring when energy does not

Optimizer cost: $f = -\ln\langle e^{-\eta E}\rangle$ with sampled_tail $\eta = \ln 20/(Q_{25} - Q_{05})$.
Toy knapsack slice, three bitstrings: ground state $E = -12$, a wrong feasible packing (trap) $E = -8$,
and an infeasible penalty wall $E = +80$. Start with the histogram mode on the trap and 37% of shots on the wall.
sampled_tail on that histogram: $Q_{05} = -12.00$, $Q_{25} = -10.76$, $\eta = 2.42$.
Boltzmann weight vs GS: trap $6.2e-05$, wall $2e-97$. Hits are read from the mode, not from $\langle H\rangle$.





At this η the wall (and almost the trap) drops out of f. Energy $\langle H\rangle = \sum p_i E_i$ still feels the +80.

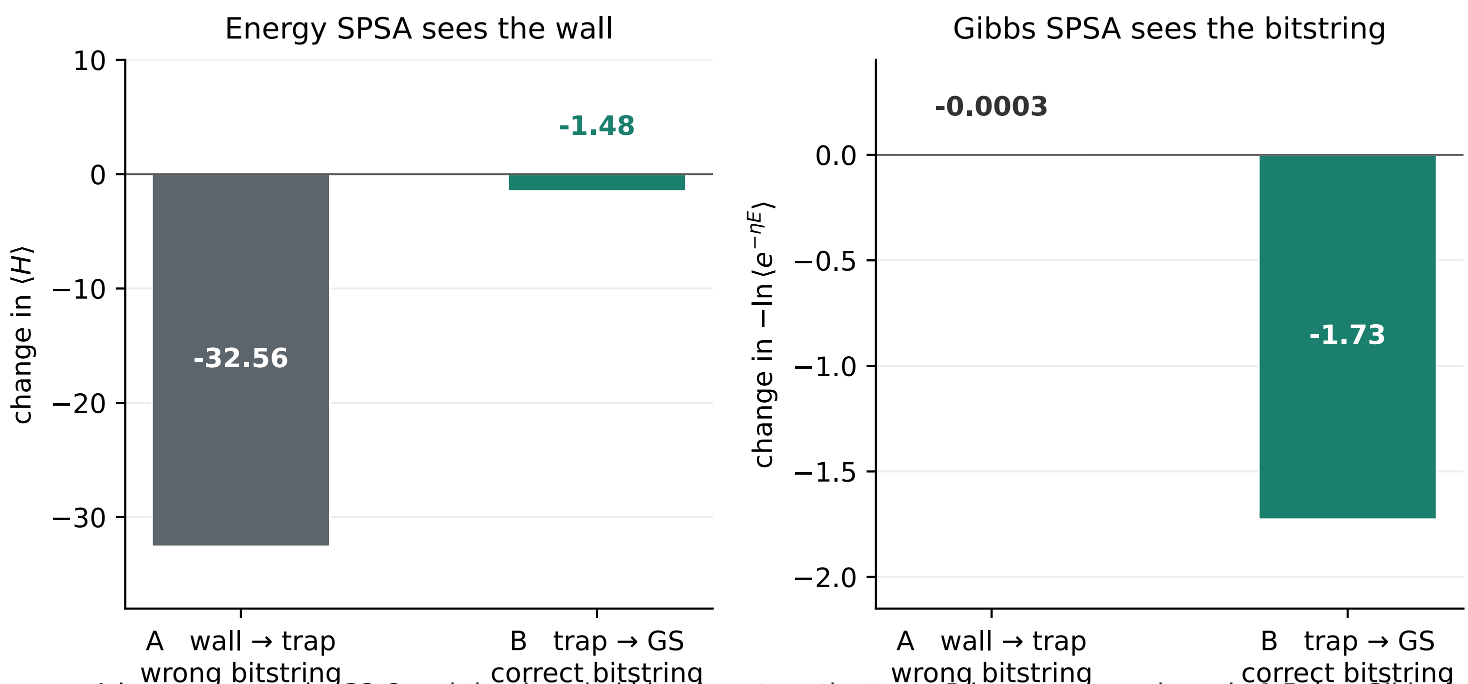
	GS	trap	wall
Boltzmann $e^{-\eta(E - E_{\mathrm{GS}})}$	1	6.18e-05	2e-97
contribution $p_i \times$ weight	0.080	3.40e-05	6e-98

Because $\langle H\rangle$ is linear, a shot on the wall costs +80 while moving trap $\rightarrow$ GS is only a gap of 4.
SPSA on energy is dominated by clearing infeasible shots, even if the most likely bitstring stays wrong.
Gibbs with this η is (up to a constant) $-\ln p_{\mathrm{GS}}$ once the wall is exponentially off.
That is why a histogram can have higher $\langle H\rangle$ and still be preferred: leftover wall shots barely enter f.

Same-size move, then a higher-energy but correct bitstring

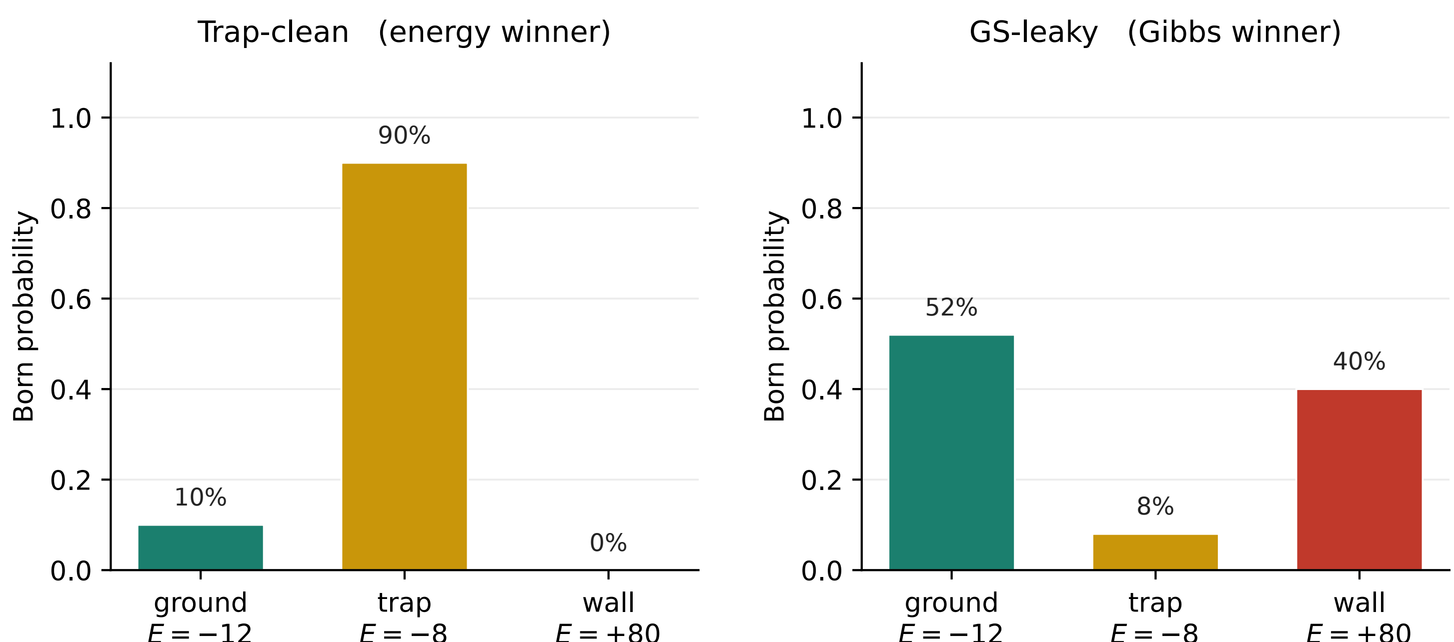
Hold η fixed (SPSA does this for both probes of a step). Move the same probability mass 0.37 two ways.

A: dump the wall onto the trap (decoder stays trap). B: dump trap mass onto the ground state (decoder becomes GS



A lowers energy by 32.6 and the decoded bitstring stays the trap. B lowers energy by only 1.5, but Gibbs by 1.73.

B has much higher energy than A (22.8 vs -8.3) and is still the Gibbs winner: the mode is now GS.



Head-to-head at the same η: trap-clean has lower energy (-8.40 vs 25.12) but the wrong bitstring.

GS-leaky still has 40% of shots on the wall, so $\langle H\rangle$ looks terrible, yet Gibbs ranks it better because the mode is GS.

	Trap-clean	GS-leaky
p (GS, trap, wall)	0.10, 0.90, 0	0.52, 0.08, 0.40
$\langle H\rangle$ (lower looks better)	-8.40	25.12
Gibbs f (lower is better)	-26.77	-28.42
most likely bitstring	trap (wrong)	GS (correct)
energy would pick	yes	no
Gibbs would pick	no	yes